

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2013-14

Paper No. MTB-201: Calculus-II (New Course)

Time: 3 Hours

Full Marks: 70

Code: BS/Sem II/2

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory.)

- (a) Give the definitions of the limit and continuity of $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ at a point (a, b) . [1+1]
- (b) Give the definitions of the derivative and partial derivatives of $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ at (a, b) . [1+1]
- (c) If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is differentiable at (a, b) , prove it is also continuous at (a, b) . [2]
- (d) Let $f(x, y) = 2x^2 + 3y^2$. Using the ε - δ definition, prove $\lim_{(x,y) \rightarrow (1,2)} f(x, y) = 14$. [2]
- (e) Find $\partial f / \partial y$ for $f(x, y) = \cos y$ at (a, b) using the limit definition. [2]
- (f) Let $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ be continuously differentiable and homogeneous of degree k . Prove $xu_x + yu_y = ku$. [2]
- (g) If $x = r \cos \theta$, $y = r \sin \theta$, show $r \frac{\partial z}{\partial r} = x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}$ and $\frac{\partial z}{\partial \theta} = x \frac{\partial z}{\partial y} - y \frac{\partial z}{\partial x}$. [1+1]
- (h) Evaluate $\int_0^{\pi/2} \int_0^{\pi/2} \sin(x + y) dx dy$. [2]
- (i) Compute the volume under $z = 1 - x - y$ over $D = [0, 1] \times [0, 1]$. [2]
- (j) Change the order of integration: $\int_0^2 \int_0^{2x} f(x, y) dy dx$. [2]
- (k) Using polar coordinates, find the area of the disc $x^2 + y^2 \leq a^2$. [2]

Question 2.

- (a) If the limit of f at (a, b) exists, show it is unique. For $f(x, y) = \frac{x^3 y}{(x^2 + y^2)^2}$, show $\lim_{(x,y) \rightarrow (0,0)} f$ does not exist. [3+3]
- (b) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = \frac{x^2 y}{x^4 + y^2}$ for $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$. Show f is not continuous at $(0, 0)$, but both partial derivatives are defined on all of $\mathbb{R}^2$. [2+4]

Question 3.

- (a) Define $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$. Show $f_{yx}(0, 0) \neq f_{xy}(0, 0)$. [6]
- (b) Let $f(x, y) = x^2y + \sin(1/x)$ for $x \neq 0$ and 0 for $x = 0$. Find f_x and f_y . Prove f_x is not continuous on the y -axis but f_y is continuous everywhere. [6]

Question 4.

- (a) Let V satisfy the Laplace equation $V_{xx} + V_{yy} = 0$. With $x = r \cos \theta$, $y = r \sin \theta$, show $V_{rr} + \frac{1}{r}V_r + \frac{1}{r^2}V_{\theta\theta} = 0$. [6]
- (b) Find the maximum and minimum of $f(x, y) = xy$ on the unit circle $x^2 + y^2 = 1$. [6]

Question 5.

- (a) Let $g(r, \theta, \phi) = (r \cos \theta \sin \phi, r \sin \theta \sin \phi, r \cos \phi)$. Compute the Jacobian matrix of g and find the determinant. [6]
- (b) State and prove Taylor's theorem for $f(x, y)$. [6]

Question 6.

- (a) For $f(x, y) = e^{x^2}$, evaluate $\iint_D f \, dx \, dy$ over $D = \{(x, y) : 0 < x < 1, 0 \leq y < 2x\}$. [6]
- (b) Change the order and evaluate $\int_0^a \int_0^{\sqrt{a^2 - x^2}} \sqrt{x^2 + y^2} \, dy \, dx$. [6]

Question 7.

- (a) For the simplex $V = \{(x, y, z) : x \geq 0, y \geq 0, z \geq 0, x+y+z \leq 1\}$, prove $\iiint_V x^{\ell-1} y^{m-1} z^{n-1} \, dV = \frac{\Gamma(\ell)\Gamma(m)\Gamma(n)}{\Gamma(\ell+m+n+1)}$. [6]
- (b) Over the region bounded by the first octant below $\left(\frac{x}{a}\right)^p + \left(\frac{y}{b}\right)^q + \left(\frac{z}{c}\right)^r = 1$, show $\iiint x^{\ell-1} y^{m-1} z^{n-1} \, dV = \frac{a^\ell b^m c^n \Gamma(\ell/p) \Gamma(m/q) \Gamma(n/r)}{pqr \Gamma(\ell/p + m/q + n/r + 1)}$. [6]

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